DSBDAL Assignment 1

Code:-

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class WordCount {
  public static class TokenizerMapper
      extends Mapper<Object, Text, Text, IntWritable>{
    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();
    public void map(Object key, Text value, Context context
                ) throws IOException, InterruptedException {
      StringTokenizer itr = new StringTokenizer(value.toString());
      while (itr.hasMoreTokens()) {
        word.set(itr.nextToken());
        context.write(word, one);
      }
    }
  }
  public static class IntSumReducer
      extends Reducer<Text,IntWritable,Text,IntWritable> {
    private IntWritable result = new IntWritable();
    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context
                    ) throws IOException, InterruptedException {
      int sum = 0;
      for (IntWritable val : values) {
        sum += val.get();
      }
      result.set(sum);
      context.write(key, result);
    }
  }
  public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
```

Output:-

```
Applications  Places  System
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
                HDFS: Number of bytes read=166
                HDFS: Number of bytes written=30
                HDFS: Number of read operations=6
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=6240
                Total time spent by all reduces in occupied slots (ms)=6177
                Total time spent by all map tasks (ms)=6240
                Total time spent by all reduce tasks (ms)=6177
                Total vcore-seconds taken by all map tasks=6240
                Total vcore-seconds taken by all reduce tasks=6177
                Total megabyte-seconds taken by all map tasks=6389760
                Total megabyte-seconds taken by all reduce tasks=6325248
        Map-Reduce Framework
                Map input records=5
                Map output records=5
                Map output bytes=60
                Map output materialized bytes=48
                Input split bytes=126
                Combine input records=5
                Combine output records=3
                Reduce input groups=3
                Reduce shuffle bytes=48
                Reduce input records=3
                Reduce output records=3
                Spilled Records=6
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=121
                CPU time spent (ms)=1240
                Physical memory (bytes) snapshot=332361728
                Virtual memory (bytes) snapshot=3007389696
                Total committed heap usage (bytes)=226365440
        Shuffle Errors
Java - WordCount/src/WordCount.java - Eclipse
[Java - WordCount/src/...  [gmail - Yahoo Search ...  cloudera@quickstart:~
```

```
Applications  Places  System
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ ls
cloudera-manager  eclipse                    Music       temp.txt
cm_api.py         enterprise-deployment.json output.txt  temp.txt~
Desktop           express-deployment.json    Pictures    Videos
Documents         kerberos                   Public      WordCount.jar
Downloads         lib                        Templates   workspace
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ cat > /home/cloudera/Processfile1.txt
Nimisha
Deepika
Nimisha
Deepike
Nimisha
^Z
[1]+  Stopped                 cat > /home/cloudera/Processfile1.txt
[cloudera@quickstart ~]$ cat /home/cloudera/Processfile1.txt
Nimisha
Deepika
Nimisha
Deepike
Nimisha
[cloudera@quickstart ~]$ hdfs dfs -ls
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 7 items
drwxr-xr-x   - hbase    supergroup          0 2026-01-15 20:28 /hbase
drwxr-xr-x   - cloudera supergroup          0 2026-01-15 20:54 /infolder
drwxr-xr-x   - cloudera supergroup          0 2026-01-15 20:59 /out1
drwxr-xr-x   - solr     solr                0 2015-06-09 03:38 /solr
drwxrwxrwx   - hdfs     supergroup          0 2026-01-15 20:13 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2015-06-09 03:38 /user
drwxr-xr-x   - hdfs     supergroup          0 2015-06-09 03:36 /var
[cloudera@quickstart ~]$ hdfs dfs -mkdir /inputfolder1
[cloudera@quickstart ~]$ hdfs dfs -put /home/cloudera/Processfile1.txt /inputfol
der1/
[cloudera@quickstart ~]$ hdfs dfs -cat /inputfolder1/Processfile1.txt
Nimisha
Deepika
Nimisha
Deepike
[Java - WordCount/src/...  [gmail - Yahoo Search ...  cloudera@quickstart:~
```

Applications Places System

Screenshot captured
You can paste the image from the clipboard.

File Edit View Search Terminal Help

```
                Map input records=5
                Map output records=5
                Map output bytes=60
                Map output materialized bytes=48
                Input split bytes=126
                Combine input records=5
                Combine output records=3
                Reduce input groups=3
                Reduce shuffle bytes=48
                Reduce input records=3
                Reduce output records=3
                Spilled Records=6
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=121
                CPU time spent (ms)=1240
                Physical memory (bytes) snapshot=332361728
                Virtual memory (bytes) snapshot=3007389696
                Total committed heap usage (bytes)=226365440
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=40
        File Output Format Counters
                Bytes Written=30
[cloudera@quickstart ~]$ hdfs dfs -ls /out12
Found 2 items
-rw-r--r--   1 cloudera supergroup          0 2026-01-20 22:47 /out12/_SUCCESS
-rw-r--r--   1 cloudera supergroup         30 2026-01-20 22:47 /out12/part-r-00000
[cloudera@quickstart ~]$ hdfs dfs -cat /out12/part-r-00000
Deepika 1
Deepike 1
Nimisha 3
[cloudera@quickstart ~]$
```

[Java - WordCount/src/... [gmail - Yahoo Search ... cloudera@quickstart:~